

LAST NAME: _____

Practice Quiz 03A — MATH 2055 — Fall 2026

Practice for Quiz 03

The following is closed book and closed note. Don't assume the grader will know what you mean. Use full sentences and provide complete and correct solutions.

Revised using GPT-6 Astra in the Codex harness.

1. Consider

$$(i) \forall x \in \mathbf{Z} \exists y \in \mathbf{Z} (y > x) \quad (ii) \exists y \in \mathbf{Z} \forall x \in \mathbf{Z} (y > x).$$

(a) State each in plain English and say whether it is true or false.

(b) Negate (ii), with no negation symbol to the left of a quantifier.